

Introduction to Algebra

AQA KS3 · Year 7 · Foundation

Name: _____ Class: _____ Date: _____

What you need to be able to do

Understand that a letter stands for a number we do not know yet. Write simple expressions, simplify them by collecting like terms, and substitute a number for a letter to find a value.

Common mistakes to avoid

- $3x$ means $3 \times x$. It is not the digits 3 and x written next to each other.
- You can only collect like terms. For example $3x + 2x = 5x$, but $3x + 2$ stays as $3x + 2$.
- x on its own means $1x$, so $x + x = 2x$ — not x^2 .
- Multiplying is different from adding: $x \times x = x^2$, but $x + x = 2x$.

Method steps

1. Step 1: Find the like terms — the ones with the same letter (and the plain numbers together).
2. Step 2: Group the like terms, keeping the $+$ or $-$ sign in front of each one.
3. Step 3: Add or subtract the number parts (the coefficients) of each group.
4. Step 4: Write the simplified expression — letters first, then the number.
5. Step 5: Check by substituting a number for the letter.

Worked example

Simplify: $4x + 3 + 2x - 1$

Step 1 — Find the like terms: $4x$ and $2x$ are like; 3 and -1 are like.

Step 2 — Group them: $(4x + 2x) + (3 - 1)$

Step 3 — Combine the x terms: $4x + 2x = 6x$

Step 4 — Combine the numbers: $3 - 1 = 2$

Answer: $6x + 2$ (check $x = 5$: $4 \times 5 + 3 + 2 \times 5 - 1 = 32$ and $6 \times 5 + 2 = 32$ ✓)

Question 1 of 8

[1 mark] ●○○○○○

Write an expression for a number, n , with 5 added to it.

How to start

1. A letter, n , stands for the number you do not know yet.
2. "5 added to it" means you do $+5$ after the n .
3. Write the letter, then the operation, then the number.

Answer: _____

Remember the method:

- 1 Find the like terms
- 2 Group them (keep each sign)
- 3 Add or subtract the number parts
- 4 Letters first, then the number
- 5 Check by substituting

Simplify.

$$a + a + a + a$$

How to start

1. Each a is one lot of a, so this is $1a + 1a + 1a + 1a$.
2. Count how many a's there are altogether and write that number in front of the a.

Answer: _____

Remember the method:

- 1 Find the like terms
- 2 Group them (keep each sign)
- 3 Add or subtract the number parts
- 4 Letters first, then the number
- 5 Check by substituting

Simplify.

$$5x + 2x - 3x$$

Answer: _____

Remember the method:

- 1 Find the like terms
- 2 Group them (keep each sign)
- 3 Add or subtract the number parts
- 4 Letters first, then the number
- 5 Check by substituting

Simplify.

$$7y + 4 + 2y + 5$$

Answer: _____

Remember the method:

- 1 Find the like terms
- 2 Group them (keep each sign)
- 3 Add or subtract the number parts
- 4 Letters first, then the number
- 5 Check by substituting

Simplify.

$$8m + 3n - 2m + 5n$$

Answer: _____

Remember the method:

- 1 Find the like terms
- 2 Group them (keep each sign)
- 3 Add or subtract the number parts
- 4 Letters first, then the number
- 5 Check by substituting

Question 6 of 8

[3 marks] ●●●○○

A pencil costs p pence.

- (a) Write an expression for the cost of 4 pencils and a 30p rubber.
- (b) Work out the cost when $p = 25$.

Answer: _____

Remember the method:

- 1 Find the like terms 2 Group them (keep each sign) 3 Add or subtract the number parts
- 4 Letters first, then the number 5 Check by substituting

Question 7 of 8

[3 marks] ●●●●○

Expand and simplify.

$$3(x + 2) + 2x$$

Answer: _____

Remember the method:

- 1 Find the like terms
- 2 Group them (keep each sign)
- 3 Add or subtract the number parts
- 4 Letters first, then the number
- 5 Check by substituting

Question 8 of 8

[3 marks] ●●●●●

A rectangle has length $(x + 3)$ and width x .

- (a) Write a simplified expression for its perimeter.
- (b) Work out the perimeter when $x = 4$.

Answer: _____

Remember the method:

- 1 Find the like terms 2 Group them (keep each sign) 3 Add or subtract the number parts
- 4 Letters first, then the number 5 Check by substituting

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

Which question did you find the hardest, and why?

One thing I now understand about using letters in maths:

One thing I will practise before the next lesson:

Teacher key

Answers

1. Q1: $n + 5$

2. Q2: $4a$

3. Q3: $4x$ ($5x + 2x - 3x = 4x$)

4. Q4: $9y + 9$ ($7y + 2y = 9y$; $4 + 5 = 9$)

5. Q5: $6m + 8n$ ($8m - 2m = 6m$; $3n + 5n = 8n$)

6. Q6: (a) $4p + 30$ (b) $4 \times 25 + 30 = 130$ pence = £1.30

7. Q7: $5x + 6$ ($3(x + 2) = 3x + 6$; $3x + 6 + 2x = 5x + 6$)

8. Q8: (a) $4x + 6$ [perimeter = $2((x + 3) + x) = 2(2x + 3) = 4x + 6$] (b) $4 \times 4 + 6 = 22$